

Point In Time Sector Seeds for Crypto Perpetual Theme Ignition

Executive summary

A usable point-in-time sector map can be built from public sources, but only if the dataset is split into three classes of evidence. The strongest class is **dated artifacts**: research reports, dated methodology documents, whitepapers, and protocol launch posts that were public at the time. The second class is **stable functional documentation**: official protocol docs that make a protocol's core function clear, but were published later or are undated current pages; these are often good enough for a stable primary sector but usually only at **medium** confidence. The weakest class is **current-only tagging systems** such as live category pages on CoinGecko, CoinMarketCap, and DefiLlama; they are useful for discovery and QA, but unsafe as sole historical truth because the live endpoints do not expose historical classification snapshots and because category assignment rules can change over time. ¹

For a sector/theme ignition strategy on crypto perpetuals, the highest-confidence reusable data are the **stable functional sectors**: currencies/payments, smart-contract platforms, rollups/L2s, oracle/data, DeFi lending/DEX, liquid staking, tokenized/RWA protocols, and AI-native protocols when AI is part of the protocol's core design rather than a later market label. Dated research frameworks from Binance Research, CoinGecko, FTSE/Grayscale, and Bitwise are especially useful because they show what categories were publicly legible at specific times. ²

The parts that are most dangerous for historical backtesting are **ecosystem tags** and **narrative tags**. CoinGecko's own methodology states that ecosystem tags such as "Ethereum Ecosystem" or "Solana Ecosystem" are auto-assigned from contract addresses, while category tags such as "AI," "DeFi," and "RWA" are manually requested market labels and subject to periodic review. That makes current CoinGecko tags a good enrichment layer, but not safe PIT truth unless you archived them yourself or can recover them from dated reports or the Wayback Machine. CoinMarketCap exposes category endpoints, but its public docs show no historical date parameter for category membership, unlike its listings-historical endpoint. DefiLlama's category pages are excellent for current DeFi protocol structure, but they are still live views unless separately archived. ³

For B1 testing of "theme ignition leader continuation," the most usable themes are: **DeFi summer 2020; alt-L1 / smart-contract rotation 2021; L2/Base and broader rollup rotation 2023–2024; AI tokens 2023–2025; Solana ecosystem and Solana memecoin waves 2023–2025; restaking 2024; TON/Telegram ecosystem 2024–2025; and RWA/tokenization 2023–2025**. These themes have reasonably clear public recognition points and liquid leader assets that were or became perpetual candidates. NFT/metaverse 2021 is usable, but more attention-driven and more vulnerable to narrow-asset distortions. Privacy rotations are real, but the evidence for broad, liquid, multi-asset breadth over 2020–2025 is weaker and exchange delistings create extra selection bias. ⁴

Source matrix

The table below separates source usefulness for **historical PIT labeling** from usefulness for **current discovery**.

source	coverage	PIT usefulness	taxonomy detail	cost / access	API / export	limitations
Messari classification system ⁵	Broad crypto taxonomy: sectors, sub-sectors, tags	Medium for stable primary sectors; low for historic reconstruction unless you had old snapshots	High	Paid features around some classifications, docs public	API + docs	Current documentation is detailed, but public evidence gathered here is mostly current-state or methodology rather than dated historical snapshots.
CoinGecko categories/ tags + research reports ⁶	Very broad token coverage, categories, ecosystem tags, quarterly/ annual reports	Low for live tags alone; medium-high when using dated CoinGecko reports	Medium	Free + paid API	API endpoints for current categories	CoinGecko says ecosystem tags are auto-assigned and category tags are manually requested market labels subject to review; no historical tag snapshot parameter is exposed in public category docs.

source	coverage	PIT usefulness	taxonomy detail	cost / access	API / export	limitations
CoinMarketCap categories ⁷	Broad market coverage and category pages	Low for PIT classification	Medium	Free + paid API	API has current category endpoints	Public docs show historical listings endpoints with a date parameter, but category endpoints do not expose a historical date parameter.
DefiLlama categories/ protocol directory ⁸	DeFi protocols and subcategories	Medium for DeFi-only functional mapping; low for narrative history unless archived	High within DeFi	Free	Public pages + API refs	Excellent for protocol categories such as lending, liquid staking, restaking, perps, RWA, AI agents; weak outside DeFi and mainly current-state unless archived.
Binance Research industry maps and sector reports ⁹	Dated market maps and theme studies	High	Medium-high	Free PDFs/ reports	Manual export	Selective universe; not an always-on API taxonomy. Very useful for “what themes were public by date.”

source	coverage	PIT usefulness	taxonomy detail	cost / access	API / export	limitations
FTSE Grayscale Crypto Sectors ¹⁰	Institutional sector framework and index rules	Medium-high from 2023 onward	High	Public methodology, investable index orientation	Factsheets / docs	Modern framework, but starts late relative to 2020–2021. Investable-universe filters can exclude many crypto names.
Bitwise Category Leaders Strategy ¹¹	Compact institutional category set at launch	Medium from April 2023 onward	Medium	Public factsheet	Manual extraction	Small, investable category set, not a full asset universe. Still useful as dated proof that categories like oracle, L2, lending, DEX, staking were public by 2023.
Official protocol docs / whitepapers / launch posts ¹²	Protocol-level primary function	High when dated and close to launch; medium when current docs only	High for primary function	Free	Usually no structured export	Best source for stable function, weak for changing market narratives, “ecosystem” tags, or sector popularity.
Wayback Machine ¹³	Archived web pages	Potentially high , but manual and sparse	Varies	Free	Archive tooling	Essential for backfilling historical page states, but coverage is uneven and extraction is manual.

Practical schema and seed sector map

The schema below is appropriate for a backtest table, but the date fields should allow **coarse strings** like `<=2020-01-01`, `2023-03`, or `2023-03-23` so the dataset does not invent day-level precision that the sources do not actually support. That is safer than silently converting vague evidence into exact timestamps. The right rule is: **primary_sector should be functional and relatively stable; secondary_tags and ecosystem_tags may change over time and must carry lower confidence unless directly evidenced by a dated source.** ¹⁴

Machine-readable seed table

The table below is a **seed**, not a full universe. It is intentionally biased toward assets that were liquid, publicly legible, and repeatedly appeared in perpetual markets or sector discussions. It should be expanded in a second pass to cover the rest of the watchlist. The `known_perp_symbols` field is a **symbol hint** for common USDT-perp naming, not proof of PIT listing eligibility; live and historical listing eligibility should be validated separately from exchange instrument masters. Bybit's linear contract docs and Binance USDⓈ-M exchange-info docs confirm the symbol/instrument lookup path, but they do not backfill historical listings by themselves. ¹⁵

asset_id	ticker	base_asset	known_perp_symbols	taxonomy_source	taxonomy_version	primary_sector
bitcoin	BTC	BTC	BTCUSDT	official whitepaper	original	currency
ethereum	ETH	ETH	ETHUSDT	official whitepaper + ethereum.org	original/current	smart_contract
solana	SOL	SOL	SOLUSDT	official docs	current docs	smart_contract

asset_id	ticker	base_asset	known_perp_symbols	taxonomy_source	taxonomy_version	primary_sector
avalanche	AVAX	AVAX	AVAXUSDT	official docs	current docs	smart_contract
cardano	ADA	ADA	ADAUSDT	official docs	current docs	smart_contract
arbitrum	ARB	ARB	ARBUSDT	official docs + dated airdrop coverage	Nitro / current	layer_2
optimism	OP	OP	OPUSDT	official docs	current docs	layer_2
starknet	STRK	STRK	STRKUSDT	official docs	current docs	layer_2

asset_id	ticker	base_asset	known_perp_symbols	taxonomy_source	taxonomy_version	primary_sector
celestia	TIA	TIA	TIAUSDT	official docs	current docs	infrastructure
chainlink	LINK	LINK	LINKUSDT	official docs + institutional sector docs	current / 2023+	utilities_and_s
pyth- network	PYTH	PYTH	PYTHUSDT	official docs	current docs	utilities_and_s
aave	AAVE	AAVE	AAVEUSDT	official docs + Bitwise	current / Apr-2023	financials

asset_id	ticker	base_asset	known_perp_symbols	taxonomy_source	taxonomy_version	primary_sector
uniswap	UNI	UNI	UNIUSDT	official docs + Bitwise	current / Apr-2023	financials
dydx	DYDX	DYDX	DYDXUSDT	official docs	current docs	financials
lido-dao	LDO	LDO	LDOUSDT	official docs + Bitwise	current / Apr-2023	financials
pendle	PENDLE	PENDLE	PENDLEUSDT	official docs	current docs	financials
ondo	ONDO	ONDO	ONDOUSDT	official launch post + docs	Jan-2023 / current	financials

asset_id	ticker	base_asset	known_perp_symbols	taxonomy_source	taxonomy_version	primary_sector
bittensor	TAO	TAO	TAOUSDT	official whitepaper + docs	original/current	artificial_intel
fetch-ai	FET	FET	FETUSDT	official docs/blog	current	artificial_intel
render	RNDR	RNDR	RNDRUSDT	official whitepaper + site	current	utilities_and_s

asset_id	ticker	base_asset	known_perp_symbols	taxonomy_source	taxonomy_version	primary_sector
toncoin	TON	TON	TONUSDT	Reuters + current ecosystem evidence	2023–2025 public theme evidence	smart_contract
bonk	BONK	BONK	BONKUSDT	official site + CoinGecko	current	consumer_application
dogwifhat	WIF	WIF	WIFUSDT	CoinGecko current page	current-only narrative page	consumer_application
pepe	PEPE	PEPE	PEPEUSDT	Reuters + market pages	2023 public frenzy	consumer_application

asset_id	ticker	base_asset	known_perp_symbols	taxonomy_source	taxonomy_version	primary_sector
ethena	ENA	ENA	ENAUUSD	official docs	current docs	financials
jito	JTO	JTO	JTOUSD	official docs	current docs	financials
worldcoin	WLD	WLD	WLDUSD	official World blog	proof_of_personhood framing	identity

Current-only or lower-confidence tags that should be annotation only

These are useful for research dashboards, but they should *not* be rankable PIT features unless you can recover their dated first appearance:

tag family	examples	why unsafe as historical truth
ecosystem tags	Solana ecosystem, Ethereum ecosystem, Base ecosystem, TON ecosystem	CoinGecko says ecosystem tags are auto-assigned from contract address relationships, which is a current-state rule, not a historical archive. ⁴³
market narrative tags	AI, RWA, meme, DePIN, GameFi on live aggregators	CoinGecko says category tags are manually requested market labels and subject to periodic review; CMC category endpoints are current-state; DefiLlama pages are live unless archived. ⁴⁴

tag family	examples	why unsafe as historical truth
backfilled cross-theme tags	labeling RNDR or WLD as “AI” back into early years; labeling LINK as “RWA” before tokenization became a public driver	The protocol may fit the later theme, but the theme was not necessarily the publicly legible trade at the time. ⁴⁵

Historical theme ignition episodes

The best use case for this strategy family is not “buy all assets in a category.” It is: detect a **publicly legible breadth impulse**, measure which liquid names within that theme are actually confirming, and then trade the strongest leader only after the theme is visible in public information and market breadth. Dated report pages from CoinGecko and Binance Research are very useful here because they mark when a narrative had become recognizable enough to be discussed as a category rather than as isolated idiosyncratic moves. ⁴⁶

Theme ignition event table

theme_event_id	theme_name	first_public_theme_ts_utc	theme_recognition_date_utc	primary_source_url
THEME_DEFI_2020	DeFi summer	2020-05-28	2020-06-15	CoinDesk on COMF distribution + Coinbase “Around the Block” on yield farming and DeFi growth ⁴⁷
THEME_ALT_L1_2021	Smart- contract platform rotation	2021-04-14	2021-09-30	CoinGecko Q1 202 report on users seeking cheaper Ethereum alternatives + CoinDesk on rotation across top L1s + Nov. 2021 AVAX/L1 performance coverage ⁴⁸
THEME_META_2021	NFT, gaming, metaverse	2021-11-01	2021-12-01	Reuters on who is building the metaverse and on Meta’s rebrand context ⁴⁹

theme_event_id	theme_name	first_public_theme_ts_utc	theme_recognition_date_utc	primary_source_url
THEME_RWA_2023	RWA and tokenization	2023-01-10	2023-07-25	Ondo tokenized Treasuries launch + Binance RWA report + Swift tokenization experiments + CoinGecko 2023 narrativization ⁵⁰
THEME_L2_2023	L2 and Base ecosystem rotation	2023-03-23	2024-01-01	Arbitrum airdrop coverage + CoinGecko 2024 report showing Base and Layer 2 among top narratives ⁵²
THEME_AI_2023	AI tokens	2023-01	2023-12-31	Chainlink/AI-adjacent market labeling aside, the clean evidence is CoinGecko 2023 report naming AI the top narrative, plus AI-native official docs for TAO and FET ⁵³
THEME_SOL_2023	Solana ecosystem revival	2023-12	2024-03-31	CoinGecko 2023 report on Solana ecosystem interest + CoinGecko 2024 Q1 report on Solana memecoin surge ⁵⁵
THEME_MEME_2024	Memecoin supercycle	2024-01	2024-12-31	CoinGecko 2024 annual report + Reuters on PEPE frenzy + CoinGecko Q1 report on Solana memecoin growth ⁵⁶

theme_event_id	theme_name	first_public_theme_ts_utc	theme_recognition_date_utc	primary_source_url
THEME_RESTAKING_2024	Restaking and liquid restaking	2024-01	2024-04-02	Binance Research restaking report + CoinGecko Q1 2024 report on EigenLayer and liquid restaking growth ⁵⁸
THEME_TON_2024	TON and Telegram ecosystem	2024-02	2025-01-21	Reuters on TON's Telegram tie-up and 2024 monetization rails + 2025 TON/Telegram mini-app exclusivity reporting ³⁶
THEME_PRIVACY_2024	Privacy / defensive rotation	2024-02	2025-12-29	Binance delisting context around XMR + Grayscale privacy preference commentary later ⁵⁹

Recommended backtesting use

A good backtest should treat this dataset as **two different products**. The first is a **rankable PIT sector map** built from `primary_sector`, `sub_sector`, and only those `secondary_tags` whose start date is supported by a dated source. The second is an **annotation layer** for narratives and ecosystem tags. Mixing them is where most lookahead mistakes happen. ⁶¹

The rankable PIT features should include:

use class	fields	rule
PIT-rankable	<code>primary_sector</code> , <code>sub_sector</code> , <code>effective_start_utc</code> , <code>effective_end_utc</code> , <code>sector_confidence</code>	Use only rows with <code>high</code> or <code>medium</code> confidence and only after <code>effective_start_utc</code> .
PIT-rankable when dated	<code>secondary_tags</code>	Only if the tag has dated public evidence. Example: <code>RWA</code> on ONDO from 2023-01-10 is usable; <code>AI</code> on RNDR from a current aggregator page is not.

use class	fields	rule
annotation only	<code>ecosystem_tags</code> , <code>current_only</code> tags, narrative popularity notes	Use for analysis and filtering, not as triggers or ranking inputs.
exclude	live aggregator categories without archive history	Exclude from historical feature generation.

The most robust first-pass implementation is:

1. Use the sector map to define a **candidate universe**.
2. Use rolling return correlation / residual clusters to verify that a theme is actually moving as a group.
3. Require a **dated public theme event** or a **breadth threshold** before enabling the strategy for that theme.
4. Rank only assets that are both **eligible perps** and **in-theme by PIT sector map**.
5. Trade the leader, not the basket.

The parts that should be excluded from B1 are the current-only ecosystem and narrative tags from live pages, any backfilled category membership recovered only from 2026-level snapshots, and assets whose theme identity changed materially but lacks a dated first-public evidence point. That includes many “AI,” “RWA,” “ecosystem,” and “memecoin” labels on aggregators. ⁴⁴

Open questions and limitations

This seed leaves some important names for phase two: DOT, NEAR, ATOM, FIL, GRT, IMX, DOGE, SHIB, XLM, BCH, ETC, ZEC, XMR, JASMY, HYPE, and TRUMP can be added with the same rules, but I have not expanded the machine-readable seed to the full watchlist here. The source base already supports doing that, especially for stable-function names and for memecoin/payment/privacy annotations. ⁶²

There is also a structural limitation in public data: the strongest “theme popularity” evidence often comes from **reports** rather than **APIs**. That means a production historical build should archive the raw pages or reports you rely on, and ideally store the excerpt, timestamp, and parser version alongside each sector-row update. Otherwise reproducibility degrades quickly as sites revise page structures and category rules. ⁶³

Backtesting-agent instructions

Use the following prompt with a backtesting agent:

Ingest the PIT sector/theme seed table as a time-bounded classification dataset. Treat `primary_sector` and `sub_sector` as rankable only when `sector_confidence` is `high` or `medium` and the simulation timestamp is within `[effective_start_utc, effective_end_utc)`. Treat `secondary_tags` as rankable only when they are not `current_only`. Treat `ecosystem_tags` and all `current_only` fields as annotation only.

For each day, build rolling co-movement clusters on the perp universe using returns, residual returns vs BTC and ETH, volume shock, OI shock, and funding shock. A theme is “ignited” only

if both conditions are met:

- a) at least one PIT-valid sector/theme bucket shows breadth across multiple assets, and
- b) the rolling cluster confirms abnormal within-theme co-movement.

Once a theme is ignited, rank eligible assets by a leader score combining relative strength, liquidity, tradability, breadth-confirmation, and distance above theme median. Go long only the strongest liquid leader after confirmation. Never backfill current-only labels into history. Never infer theme validity from present-day category pages unless the dataset contains a dated start record.

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